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KILAMBI RESEARCH WORKING PAPER No. KR-WP-2026-11

Can the AI Buildout Pay for Itself? Demand Absorbability, Price Declines, and the Timing Mismatch in AI Infrastructure Investment

a factor of five once definition classes are separated: 2030 global datacenter capacity estimates range from 108 GW (IEA, consumption-derived) to 490.7 GW (TrendForce, nameplate; scope unverified), and pipeline figures collapse by 79–89% between interconnection requests and build-ready capacity. The Jevons test finds no support for price-driven backfire: at the only causal elasticity estimate available ($|e| = 0.5$), efficiency gains predict required capacity shrinking about 55% per year, against observed US IT capacity growth of 53% per year over 2021–2024, so current capacity growth is driven by adoption and new workloads rather than the price channel. The scenario model implies each \$1T of cumulative investment requires about \$250B per year of sustained end-market AI revenue in the base case (\$2.50T per year at the \$10T base case, supporting 154 GW). A timing-mismatch model shows the central stranding mechanism: hardware is 90% of a 30-year facility’s lifetime gross capex but only 60% of first cost, so long-dated revenue commitments made at today’s prices overpay by 87% in present value if compute prices fall 15% per year. Analogues (1990s telecom, merchant power, Chinese solar) show technology can succeed while original investors lose: Global Crossing debt recovered about 5.5 cents on the dollar and Calpine equity fell about 99.4%. Current market signals are layered: contracted power, interconnection, and preleased capacity remain tight, while marketplace GPU prices, secondary hardware, and uncontracted projects are softening; no verified tenants are walking away from energized capacity. The paper issues no binary bubble verdict.

Keywords: AI infrastructure; datacenters; capital expenditure; demand forecasting; Jevons paradox; stranded assets; GPU pricing

JEL Classification: D22, D24, G32, L63, L94, O33, Q41

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1 Introduction

Hyperscaler capital expenditure has grown at 31.6% per year from 2015 to 2025, and announced AI infrastructure programs now run into the trillions of dollars (Kilambi Research, 2026a). Two companion papers frame this one. KR-WP-2026-06 (2026a), “Powering the Boom,” documents the buildout, the efficiency trends (FLOPS/W doubling roughly every four years, training compute doubling every 6.2 months), and the physical tightness of the current market. KR-WP-2026-10 (2026b), “AI Capex and the Corporate Credit Market,” analyzes how the buildout is financed and what it implies for credit. This paper, the third in the sequence, asks the demand-side question: can end-market demand absorb the planned capacity at prices that clear investors’ hurdle rates?

The question matters because the two sides of the investment case are evaluated with different evidence standards. Supply plans are announced in press releases and counted in gigawatts; demand is modelled, inferred from workload projections, or simply assumed. When the two are compared without reconciling definitions, the comparison is meaningless: a 490 GW nameplate figure and a 108 GW consumption-derived figure are not two estimates of the same object. This paper’s first contribution is therefore definitional. It audits 52 published forecasts, assigns each to a definition class (modelled demand, announced pipeline, under construction, contracted, energized, nameplate, utilized), and shows that dispersion across forecasts is driven more by definition than by disagreement about the world.

The second contribution is a required-revenue model: a per-GW hurdle that translates capex, financing, energy, and operating assumptions into the annual revenue each GW of facility capacity must generate. The central estimate is \$3.55B per GW-year, with bounds of \$2.39B to \$4.43B driven by capex uncertainty of \$8M to \$15M per MW-IT (MODELLED, B). At the September 2026 marketplace price floor of \$1.79 per GPU-hour (OBSERVED, C), that hurdle implies 32.6% paid utilization; at a three-year accelerator economic life rather than five, the break-even rises to 45%.

The third contribution is a Jevons test. The claim that efficiency gains will expand AI compute demand enough to absorb any capacity (the “Jevons” argument) is evaluated against the identity relating efficiency gains to required physical capacity at a given elasticity of demand. At the only causal elasticity estimate available, $|e| = 0.5$, efficiency predicts required capacity *shrinking* about 55% per year using the token-price composite, against observed US nominal IT capacity growth of 53.1% per year over 2021–2024 (OBSERVED, B). Reconciliation requires an outward adoption shift of roughly 3.4x per year. The price channel does not support Jevons backfire; current capacity growth is driven by adoption, quality expansion, and new workloads.

The fourth contribution maps cumulative 2026–2035 investment scenarios of \$5T, \$10T, and \$15T into sustained end-market revenue requirements. Each \$1T of cumulative investment requires about \$250B per year of sustained end-market AI revenue in the base case; the \$10T base case implies \$2.50T per year supporting 154 GW (MODELLED, B). An efficiency/disruption scenario (GPU prices at \$1.25/hour, 45% utilization, four-year life) is unsustainable at every investment level because required utilization reaches 158%.

The fifth contribution is a timing-mismatch model and three historical analogues. A 30-year facility contains roughly six accelerator generations at a five-year economic life, and hardware is 90% of lifetime gross capex but only 60% of first cost (MODELLED, B). A 10-year revenue commitment signed at \$2.50 per GPU-hour overpays by 86.8% in present value if market pricing falls 15% per year (MODELLED, B). Telecom (Global Crossing debt recovery of about 5.5 cents per dollar, equity effectively wiped out), merchant power (Calpine equity down about 99.4%, senior debt near full

recovery), and Chinese solar (polysilicon prices down about 96%, Suntech defaulted) each show the same pattern: the technology can succeed while the original investors lose. The paper therefore issues no binary bubble verdict; the relevant question is who bears the repricing, not whether AI as a technology succeeds.

Current market evidence, coded from 41 dated observations, supports a layered verdict: contracted power, interconnection, land, and preleased capacity remain tight (20 tight readings), while marketplace GPU prices, secondary hardware, some credit spreads, and uncontracted projects are softening (12 soft, 9 mixed). No verified tenants are walking away from energized capacity. Direct AI-fleet utilization remains undisclosed by every operator, which is the central missing datum of the entire debate.

The paper proceeds as follows. Section 2 describes the institutional setting: how AI datacenters are built, priced, and contracted. Section 3 reviews the related literature. Section 4 describes the four datasets. Section 5 details the methodology. Section 6 presents results. Section 7 states limitations and robustness checks, including an explicit falsifiers subsection. Section 8 concludes. A data appendix documents variable definitions and reproduction files.

2 Institutional background

An AI datacenter converts two inputs, power and accelerators, into rentable compute. The facility shell (land, building, power distribution, cooling) has an economic life of 20 to 30 years; the accelerators inside it have an economic life of roughly three to six years, with five years the central assumption used in this paper (ASSUMED, B). This mismatch between the facility life and the hardware life is the structural fact from which the paper’s timing analysis follows.

Capital cost is conventionally quoted per MW of IT load. Industry figures cluster between \$8M and \$15M per MW-IT for AI-ready facilities, with \$12M per MW-IT the central assumption here (ASSUMED, B); the range reflects differences in power density, cooling design, and how much of the electrical and mechanical infrastructure is counted. At a power usage effectiveness (PUE) of 1.15 (ASSUMED, B), each GW of facility power supports about 870 MW of IT load. Energy cost is taken at \$0.06 per kWh (ASSUMED, B), and non-energy operating cost at 5% of capex per year (ASSUMED, B). These assumptions are deliberately middle-of-the-road; Section 5.2 varies the capex input to produce the \$2.39B to \$4.43B bounds on required revenue.

Compute is sold in tiers. Hyperscaler on-demand pricing for H100 SXM-class GPUs is about \$5.38 per GPU-hour; neocloud on-demand is about \$3.99; the marketplace floor is about \$1.79; and spot pricing is about \$0.98 (OBSERVED, C), all September 2026 snapshots. The tier structure matters because the break-even utilization of a facility depends on which price the marginal GPU-hour clears at: the same \$3.55B per GW-year hurdle implies 10.9% paid utilization at hyperscaler pricing but 59.5% at spot pricing. Price declines are the norm in this market: token prices have fallen at a composite rate of about 161% per year in log terms (Epoch AI, 2026; Gundlach et al., 2026), and hardware efficiency (FLOPS/W) improves about 19% per year (Epoch AI, 2023; Kilambi Research, 2026a).

Contracting practice layers long-dated commitments over short-dated assets. Developers sign 10- to 15-year leases and power contracts for the facility while filling it with accelerators that will be replaced multiple times over the lease term. Financing follows the project-finance pattern described by Esty (2004): 60% debt at 6% and 40% equity at 12% are the central assumptions here (ASSUMED,

B), consistent with investment-grade hyperscaler balance sheets and the structured vehicles analyzed in KR-WP-2026-10 (Kilambi Research, 2026b). Electricity procurement uses long-term contracts whose valuation follows the forward-pricing logic of Bessembinder and Lemmon (2002) and the PPA structures analyzed by Qorbanian et al. (2024); corporate renewable PPAs are a partial template for how AI load is contracting power, though AI load’s flat 24/7 profile differs from intermittent renewables.

Interconnection is the binding physical constraint in many markets. Queue positions, not chip supply, determine when load can energize: ERCOT held about 410 GW of large-load interconnection requests against 45.4 GW that SemiAnalysis tracks as commercially real (OBSERVED, B), and CAISO’s large-load issue paper documents the same queue dynamics in California (California Independent System Operator, 2026). Under-construction capacity in North America is about 66 GW, of which 95% is precommitted (OBSERVED, B). Datacenter energy demand is surveyed nationally by Shehabi et al. (2024), and load-flexibility options are summarized by Kirchstetter et al. (2025); neither source publishes AI-fleet utilization, which remains the central missing datum.

The financing layer is analyzed in KR-WP-2026-10 (Kilambi Research, 2026b) and is not repeated here. This paper takes the capital structure as an input to the hurdle rate and focuses on whether end-market demand can service it.

3 Related literature

This paper sits at the intersection of four literatures: infrastructure forecasting, electricity demand analysis, stranded-asset economics, and the economics of AI.

Forecasting bias in megaprojects. Flyvbjerg et al. (2002) document systematic cost underestimation and benefit overestimation across transportation megaprojects, and Flyvbjerg (2021) extends the finding across project types and decades. The mechanism, optimism bias compounded by strategic misrepresentation, applies directly to announced AI capacity pipelines: interconnection requests and press-release gigawatts are the benefits side of the ledger, and they are overstated by construction. The forecast audit in Section 5.1 is designed around this concern, separating definition classes before comparing numbers.

Electricity demand forecasting errors. Craig et al. (2002) show that long-run US electricity forecasts have historically overestimated demand, in part because they extrapolated the growth rates of the recent past through structural breaks. The AI load forecasts audited here face the same risk in reverse: several extrapolate the 2021–2024 capacity growth rate forward without modelling the adoption curve that produced it. The Jevons test in Section 5.3 separates the price channel from the adoption channel precisely to avoid repeating that error.

Stranded assets and irreversible investment. Ansar et al. (2013) and Mercure et al. (2018) analyze how long-lived capital becomes stranded when demand or technology paths diverge from expectations. Dixit and Pindyck (1994) provide the real-options logic: irreversible investment under uncertainty carries an option value of waiting, so committed capital that cannot be redeployed is exposed to the full downside of a demand shortfall. AI datacenters are a clean application: the facility shell is irreversible and the accelerators depreciate faster than the contracts written against them. The timing-mismatch model in Section 5.5 formalizes the exposure.

Merchant power and telecom as analogues. Joskow (2006) analyzes the US merchant-power experience, where overbuilding against optimistic demand forecasts produced the Calpine restructuring;

Organisation for Economic Co-operation and Development (2003) surveys generation investment under electricity market liberalization, the institutional setting that produced it. The telecom overbuild of the late 1990s, documented in the trade press and restructuring literature, provides the closest demand-side analogue: real demand growth combined with a large overshoot in capacity investment. Section 5.7 uses these episodes as disciplined analogues rather than rhetorical ones, comparing investor-level returns rather than asserting that “this time is the same.”

Learning curves and efficiency. Irwin and Klenow (1994) estimate learning-by-doing in semiconductors, the canonical precedent for accelerator cost declines. Strubell et al. (2019) quantify the energy cost of training large language models, establishing the physical link between model scale and power demand that underlies every workload-based capacity forecast. The Jevons analysis in Section 5.3 uses hardware efficiency (19% per year, Epoch AI 2023) and the token-price composite (161% per year in log terms, Epoch AI 2026; Gundlach et al. 2026) as its two efficiency series.

AI economics. Acemoglu (2025) evaluates the macroeconomic stakes of AI adoption, providing the demand-side context in which the scenario revenue requirements of Section 6.4 must be judged: \$2.50T per year of sustained end-market AI revenue is a claim about the size of the AI economy, not just about datacenters. Cahn (2024) poses the industry version of the absorbability question (“AI’s \$600B question”), arguing from revenue arithmetic that the buildout requires end-market revenue growth that has not yet materialized; this paper formalizes that arithmetic. Goldman Sachs Global Investment Research (2024) provide a workload-demand forecast (110 GW) that enters the forecast audit as one definition class among several. Alphabet Inc. (2024) and Meta Platforms, Inc. (2025) disclose the hyperscaler capex series that KR-WP-2026-06 uses to establish the 31.6% annual growth rate; this paper takes that series as given and does not re-estimate it.

Cloud pricing and credit. Ben-Yehuda et al. (2013) analyze EC2 spot pricing, the institutional precedent for the marketplace and spot GPU tiers in Section 2; the elasticity grid in Section 5.3 includes an EC2-based elasticity of 1.20 alongside the causal estimate of 0.5. Fitch Ratings (2025) provide the credit-rating view of datacenter and AI-infrastructure obligors that informs the market-evidence coding in Section 5.8; the full credit analysis is in KR-WP-2026-10 (Kilambi Research, 2026b). Finvestly (2026) compile the stalled-project data (over \$150B of US datacenter projects stalled or cancelled) that enters the market-evidence dataset as softness readings. Offutt et al. (2026) survey datacenter energy consumption for policymakers, a useful check on the energy-cost assumptions in Section 5.2.

The gap this paper fills is the absence of a unified demand-side accounting: forecasts are not reconciled by definition, hurdle rates are not translated into utilization requirements, the Jevons claim is not tested against an elasticity identity, and the timing mismatch between asset lives and contract lives is asserted rather than modelled. Each of those steps is taken here, with the provenance and uncertainty of every input stated alongside the number.

4 Data

Four datasets were compiled in September 2026. All are stored as CSV files with provenance and evidence-grade columns; the data appendix defines every variable.

Forecast audit dataset (52 rows). Each row records one published forecast of AI datacenter capacity, investment, or a closely related quantity: publisher, publication date, target year, headline figure, units, definition class, and evidence grade. Definition classes are: modelled demand, announced

pipeline, under construction, contracted, energized, nameplate, and utilized capacity. No source forecasts utilized compute directly; that column is empty across all 52 rows, which is itself a finding. Major scope examples: McKinsey’s \$6.7T through 2030 *including hardware* (modelled investment, not capacity), PwC’s \$31.6T through 2050 *including repeated hardware refreshes*, JLL’s 66 GW under construction with 95% precommitted, and ERCOT’s roughly 410 GW of interconnection requests against 45.4 GW that SemiAnalysis tracks as commercially real. Several items are retained with explicit weakness flags rather than dropped: BCG’s “100–130 GWh” unit problem, Morgan Stanley’s “126 GW annually” ambiguity, Dell’Oro’s \$3T annual-versus-cumulative ambiguity, TrendForce’s 490.7 GW (scope unverified), and paywalled or secondary-only items from McKinsey, BNEF, Bain, PwC, SemiAnalysis, and JPMorgan. There is no dedicated China forecast workstream; Chinese capacity enters only through global aggregates.

Demand-chain dataset. A structured chain from power procurement through GPU rental pricing to end-market revenue, with each link carrying a value, units, year, provenance, and evidence grade. It includes the funnel ratios (ERCOT 410 GW to 45.4 GW; DFW 3.7 GW planned-not-begun to 0.77 GW under construction; North America 66 GW under construction to 62.7 GW precommitted), the required-revenue components, the GPU tier price snapshots, the Jevons elasticity grid, the scenario-model parameters, and the timing-mismatch parameters.

Market-evidence dataset (41 dated rows). Each row records one dated market observation from 2025–2026, coded into one of seven categories (vacancy and preleasing; commitments and orders; credit and finance; cancellations and delays; pricing and secondary; interconnection and queues; utilization and equipment) and one of three verdicts (tight, mixed, soft). Counts: 20 tight, 12 soft, 9 mixed. The dataset is a structured reading of trade press, company disclosures, and rating-agency notes; it is not a random sample of news, and the coding reflects the author’s judgment. The unverified NVIDIA \$99B program, Broadcom \$29B backstop, AMD \$4.1B guarantee, and Goldman 353/381 bp spread figures are excluded from the dataset and are not used as verified claims anywhere in the paper.

Scenario-model dataset. Parameters and outputs of the investment-to-revenue mapping described in Section 5.4, including the \$5T/\$10T/\$15T investment levels, the base/high-demand/efficiency-disruption parameter sets, and the resulting revenue requirements and implied GW.

Table 1 reports summary statistics for the quantitative inputs. Every figure and table in the paper is generated from these datasets by scripts stored alongside them; no number in the text is hand-typed from a secondary source without a dataset row behind it.

5 Methodology

5.1 Forecast audit and definition classes

Each of the 52 forecast rows is assigned to exactly one definition class, defined as follows. *Modelled demand* is a quantity produced by an economic or workload model (e.g., McKinsey’s 219 GW, Goldman’s 110 GW workload demand). *Announced pipeline* is capacity in press releases, MOUs, or developer announcements. *Under construction* is capacity with steel in the ground. *Contracted* is capacity with signed offtake. *Energized* is capacity with power flowing. *Nameplate* is the sum of announced facility ratings regardless of stage. *Utilized* is capacity actually running paid workloads; no source forecasts this.

Table 1. Summary statistics for the paper’s quantitative inputs. Evidence grades use the house vocabulary: (OBSERVED, A) is directly measured, (RECONSTRUCTED, B) is derived from disclosed figures, (MODELLED, B) is model output from stated inputs, (ASSUMED, B) is an author assumption. Ranges are input uncertainty, not confidence intervals.

Input	Central value	Range or note	Evidence grade
Required revenue per GW-year	\$3.55B	\$2.39B–\$4.43B (capex \$8–15M/MW-IT)	(MODELLED, B)
CoreWeave Q2 annualized revenue per GW	\$6.87B	guided YE run rate \$11.5B	(RECONSTRUCTED, B)
Marketplace floor, H100 (Sep 2026)	\$1.79/GPU-hr	snapshots vary by week	(OBSERVED, A)
Neocloud on-demand, H100 (Sep 2026)	\$3.99/GPU-hr		(OBSERVED, A)
Hyperscaler on-demand, H100 (Sep 2026)	\$5.38/GPU-hr		(OBSERVED, A)
Spot floor, H100 (Sep 2026)	\$0.98/GPU-hr		(OBSERVED, A)
Break-even paid utilization (floor price)	32.6%	45% at 3-yr hardware life	(MODELLED, B)
2030 forecasts, definitional range	108–490.7 GW	52 rows, 7 definition classes	(OBSERVED, A)
ERCOT requests to build-ready	11% survives	410 GW to 45.4 GW	(OBSERVED, A)
US nominal IT capacity growth, 2021–2024	53.1%/yr	price channel predicts –55.3%/yr	(OBSERVED, A)
Market-evidence rows	41	20 tight / 9 mixed / 12 soft	(OBSERVED, A)
Accelerator share of 30-yr lifetime capex	90%	vs 60% of first cost	(MODELLED, B)
10-yr lease overpayment at –15%/yr prices	86.8%	present value vs market	(MODELLED, B)

The audit’s rule is that figures may only be compared within a definition class. The 2030 dispersion chart (Figure 2) therefore colors bars by definition class rather than ranking publishers. Weak items are retained with flags: BCG’s “100–130 GWh” (a unit error in the source, retained as flagged rather than silently converted), Morgan Stanley’s “126 GW annually” (ambiguous whether annual additions or cumulative), Dell’Oro’s \$3T (ambiguous annual versus cumulative), and TrendForce’s 490.7 GW (scope unverified, retained as the nameplate-class observation it is). Paywalled or secondary-only items are flagged as such and given no more weight than their provenance supports.

The funnel analysis (Figure 1) uses three paired observations where both the starting figure and the surviving figure are documented: ERCOT interconnection requests (410 GW) to SemiAnalysis-tracked commercially real capacity (45.4 GW); DFW planned-not-begun (3.7 GW) to under construction (0.77 GW); and North America under construction (66 GW) to precommitted (62.7 GW). These are three different markets and stages, not a single funnel; the point is the attrition rate at each transition, not a combined figure.

5.2 Required-revenue model

The required revenue per GW of facility power per year is the annual revenue that clears the weighted hurdle on the capital invested, plus energy and operating costs. Inputs: capex of \$12M per MW-IT (bounds \$8M and \$15M), five-year straight-line depreciation of the accelerator-heavy capex, 60% debt at 6%, 40% equity at 12%, PUE of 1.15, energy at \$0.06 per kWh, and opex at 5% of capex per year. All inputs are (ASSUMED, B) except the financing rates, which reflect investment-grade hyperscaler funding conditions documented in KR-WP-2026-10 (Kilambi Research, 2026b).

Depreciation dominates: \$2.40M per MW-IT per year, or \$2.09B per GW of facility power per year after the PUE conversion. Interest is \$0.37B, the equity charge \$0.50B, energy \$0.07B, and opex \$0.52B, summing to \$3.55B per GW-year. Varying only the capex input from \$8M to \$15M per

MW-IT gives the \$2.39B to \$4.43B bounds; the financing, energy, and opex assumptions are held fixed, so the bounds understate total uncertainty.

Break-even paid utilization at a GPU price p per hour is the hurdle divided by the revenue a GW generates at 100% paid utilization at that price. A GW of facility power at PUE 1.15 supports about 870 MW of IT load; at 700W per H100-class GPU including overhead, that is roughly 1.24 million GPUs, generating $p \times 1.24\text{M} \times 8760$ per year at full paid utilization. At \$1.79 per GPU-hour (marketplace floor), full utilization yields about \$10.9B per GW-year, so the \$3.55B hurdle implies 32.6% paid utilization. The paper reports the same calculation at \$3.99 (neocloud, 14.6%), \$5.38 (hyperscaler, 10.9%), and \$0.98 (spot, 59.5%). Shortening the accelerator economic life to three years raises depreciation and lifts the marketplace-floor break-even to 45%.

5.3 Jevons test

The Jevons claim is that efficiency gains reduce the effective price of compute and thereby expand demand enough to absorb any capacity. The test uses the identity

$$\frac{d \ln K}{d \ln e} = |e| - 1, \quad (1)$$

where K is required physical capacity, e is the efficiency index, and $|e|$ is the price elasticity of demand. A 1% efficiency gain changes required capacity by $(|e| - 1)\%$, converted to percent-per-year via $\exp(x) - 1$ for the reported figures. Two efficiency series are used: hardware efficiency at 19% per year (Epoch AI, 2023) and the token-price composite at 161% per year in log terms (Epoch AI, 2026; Gundlach et al., 2026). The elasticity grid uses three points: $|e| = 0.5$, the only causal estimate available (Musolff 2025); $|e| = 1.20$, an EC2-based estimate (Jiang 2013); and $|e| = 2.25$, a naive LLM-query elasticity. These are illustrative grid points from the literature, not estimates produced by this paper.

The test compares the predicted required-capacity path against observed US nominal IT capacity growth of 53.1% per year over 2021–2024 (OBSERVED, B). If the prediction undershoots the observation, the residual is attributed to outward adoption shifts (new workloads, quality expansion), computed as the ratio of observed to predicted growth factors. The identifying assumption is that the elasticity is stable over the comparison window; if demand has become more elastic over time, the price channel explains more of the growth than the central calculation allows.

5.4 Scenario model

The scenario model maps cumulative 2026–2035 investment into required sustained end-market AI revenue per year. Investment levels are \$5T, \$10T, and \$15T. Three parameter sets are used. The base case uses the required-revenue model’s central parameters. The high-demand case assumes stronger pricing power (lower required revenue per GW). The efficiency/disruption case assumes GPU prices fall to \$1.25 per GPU-hour, paid utilization of 45%, and a four-year hardware life.

For each investment level and parameter set, the model computes the GW implied by the investment at the assumed capex per GW and the annual revenue required to service it. Sustainability is defined as required paid utilization at or below 100%; the efficiency/disruption case fails this test at every investment level (required utilization of 158%) and is therefore marked not sustainable. The model is deliberately simple: it assumes the investment is deployed uniformly, ignores financing

dynamics (covered in KR-WP-2026-10), and treats end-market revenue as the residual claimant on the full capital charge.

5.5 Timing-mismatch model

Two calculations formalize the asset-life mismatch. First, lifetime capex accounting: a 30-year facility at \$12M per MW-IT first cost, with accelerator cohorts replaced every five years at constant real cost, implies six hardware generations; hardware is 90% of 30-year lifetime gross capex but 60% of first cost. Second, the lease-overpayment calculation: a 10-year revenue commitment fixed at \$2.50 per GPU-hour, against a market price that falls 15% per year, has a present value 86.8% above the market-price path. The 15% decline rate is illustrative; the qualitative result (long-dated fixed commitments overpay when prices fall) holds for any positive decline rate.

5.6 Utilized-compute forecast model

The forecast audit found that no publisher forecasts utilized compute: the series the absorbability question turns on is the missing datum. This subsection constructs it as scenario-conditioned paths, not as a prediction. Three capacity anchors are used, all in GW of AI facility power, global, 2026–2035. The central anchor is this paper’s own scenario-model path: 50 GW in 2026, rising linearly to the \$10T base-case end-state stock of 154 GW in 2035. The 2026 starting stock is triangulated from EPRI’s US 2024 nominal IT figures scaled to global and is the weakest input; it is graded accordingly. The low anchor is realization-graded: EPRI’s US low scenario of 56 GW total datacenter capacity in 2030, scaled to global at 2.5x and an AI share of 55%, giving 77 GW of AI capacity in 2030 and 91 GW in 2035. The high anchor follows forecaster demand: McKinsey’s AI demand series, 44 GW in 2025 to 156 GW in 2030, extended at 12% per year to 275 GW in 2035. All three anchors are (MODELLED, C).

Three paid-utilization paths are applied to the central anchor; the capacity path is held scenario-invariant because the build is committed capex, which is the paper’s maintained assumption about the pipeline. Paid utilization is the share of GPU-hours sold at or above the scenario’s realized price, distinct from silicon utilization. All three paths start at 45% in 2026, inside the 40–70% bracket spanned by the available hyperscale and lab utilization proxies. The base path rises to 58% by 2029 as adoption continues, then declines to 46% by 2035 as efficiency gains bite. The high-adoption path rises to 70% and holds, corresponding to the 3.4x-per-year outward demand shift the Jevons test requires. The efficiency-disruption path falls to 10% by 2035 as accelerator price-performance outruns demand growth. Realized prices are \$1.79 per GPU-hour flat in the base case (the September 2026 marketplace floor), \$2.50 flat in high adoption, and \$1.79 declining 15% per year in the disruption case. Utilization paths are (MODELLED, C) throughout.

Utilized compute is reported in H100-equivalent EFLOP/s at 2026 silicon: 696 H100-class GPUs per MW of facility power at 989 TFLOPS (FP16 dense) gives 688 EFLOP/s per GW at full utilization. The conversion is fixed at 2026 silicon; hardware efficiency gains would raise effective compute per GW over time, which widens the disruption scenario’s oversupply, so the fixed-silicon convention is conservative for the oversupply case. The hurdle lines in Figure 10 are the required-revenue model’s break-evens at the floor price: 32.6% at a five-year accelerator life and 45% at a three-year life (MODELLED, B).

5.7 Analogue method

Three historical episodes are used as disciplined analogues: the late-1990s telecom overbuild (Global Crossing), the early-2000s US merchant-power overbuild (Calpine), and the late-2000s Chinese solar manufacturing overbuild (Suntech, polysilicon). The selection criterion is structural similarity: large irreversible capacity investment, against demand forecasts that were directionally right but quantitatively overshot, with a technology-cost decline that repriced the asset base. The comparison is at the investor-return level (debt recovery rates, equity drawdowns, input price declines), not at the level of asserting the episodes are identical. Each analogue’s outcome is reported with the caveat that the AI case differs in demand visibility and contract structure.

5.8 Market-evidence coding

The 41 market-evidence rows are coded by the author into seven categories and three verdicts as described in Section 4. The coding rule for the verdict is the direction of the observation relative to a balanced market: rising prices, full preleasing, and queue growth code as tight; falling marketplace prices, cancellations, and widening spreads code as soft; observations with offsetting elements code as mixed. Figure 8 reports the category-by-verdict matrix. The dataset is not a random sample, and the counts should be read as a structured reading of the evidence, not as a statistical estimate of market conditions.

6 Results

6.1 Forecast dispersion

Figure 1 shows the funnel attrition: of every 100 units of capacity at each pipeline stage, about 11 survive from interconnection request to build-ready (ERCOT), about 21 from planned to under construction (DFW), and about 95 from under construction to precommitted (North America). The first two transitions are where announced gigawatts die; by the time steel is in the ground, the capacity is nearly all spoken for. The implication for the debate is that pipeline figures should be discounted by roughly an order of magnitude before they are compared with demand forecasts, a discount the press-release numbers do not carry on their face.

Figure 2 shows the 2030 dispersion once definition classes are separated. The range runs from 108 GW (IEA, consumption-derived from 945 TWh at 100% capacity factor) to 490.7 GW (TrendForce, nameplate; scope unverified). Between them sit Goldman Sachs at 110 GW (workload demand, Goldman Sachs Global Investment Research 2024), PwC at 127 GW (installed), BCG at 198 GW (supply-constrained), JLL at 200 GW (blended outlook), Bain at 200 GW (AI workload requirement), and McKinsey at 219 GW (modelled demand). The ordering is almost entirely explained by definition class: nameplate exceeds modelled demand, which exceeds supply-constrained and installed figures, which exceed consumption-derived figures. Publisher disagreement about the world is second-order relative to publisher disagreement about what is being counted.

Table 2 collects the forecast-audit highlights: the figures most often quoted in the debate, with the scope qualification that determines what each one actually measures.

No source in the 52-row audit forecasts utilized compute. The entire forecasting industry is therefore forecasting supply-side and modelled-demand quantities, never the paid-utilization quantity on

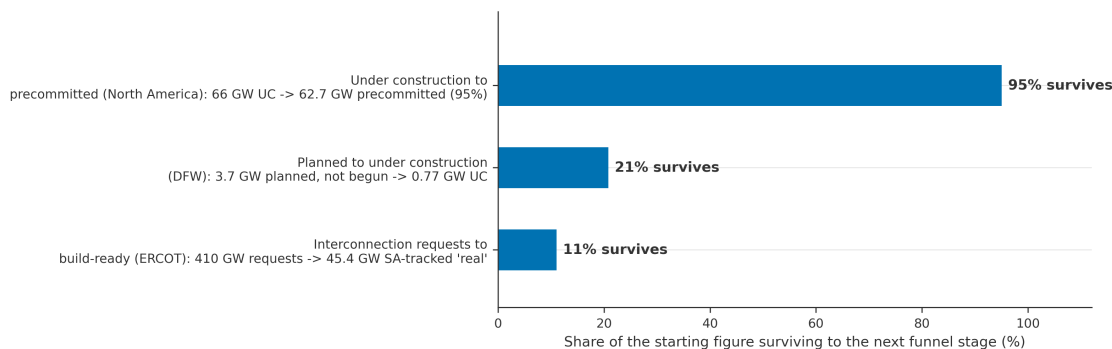


Figure 1. Pipeline attrition between capacity-definition stages. Each bar is the share of the starting figure that survives to the next stage: 410 GW of ERCOT interconnection requests to 45.4 GW SemiAnalysis-tracked commercially real capacity (11%); 3.7 GW planned-not-begun to 0.77 GW under construction in DFW (21%); 66 GW under construction to 62.7 GW precommitted in North America (95%). The three pairs are different markets and stages, not a single funnel.

which the investment case depends. That absence is the empirical counterpart to the undisclosed AI-fleet utilization noted in the introduction.

6.2 The capital hurdle

Figure 3 decomposes the \$3.55B per GW-year hurdle (MODELLED, B). Depreciation is \$2.09B, the interest charge \$0.37B, the equity charge \$0.50B, energy \$0.07B, and opex \$0.52B. The whisker shows the \$2.39B to \$4.43B bounds from varying capex between \$8M and \$15M per MW-IT; financing, energy, and opex assumptions are held fixed, so the bounds understate total uncertainty. Two market reference points are shown: CoreWeave’s Q2 annualized revenue of \$6.87B per GW-year (RECONSTRUCTED, B), which clears the hurdle with margin, and its guided year-end run rate of \$11.5B (RECONSTRUCTED, B). Both are company-reported figures reconstructed to a per-GW basis; the reconstruction assumes the reported revenue maps to the disclosed capacity base, which the company does not state directly.

Figure 4 translates the hurdle into paid-utilization terms at each rental tier. At hyperscaler on-demand pricing (\$5.38 per GPU-hour), the hurdle implies 10.9% paid utilization; at neocloud pricing (\$3.99), 14.6%; at the marketplace floor (\$1.79), 32.6%; at spot (\$0.98), 59.5% (MODELLED, B). The dotted line marks the break-even GPU price of \$0.58 per hour at 100% paid utilization. The tier structure is the whole story: the same facility is comfortably economic at hyperscaler prices and marginal at spot prices, so the absorbability question is really a question about which tier the marginal GPU-hour clears at as capacity grows. If accelerator economic life is three years rather than five, depreciation rises and the marketplace-floor break-even moves from 32.6% to 45%.

The hurdle’s sensitivity is asymmetric. It is most sensitive to the capex input and the hardware life, the two quantities with the widest honest ranges; it is least sensitive to energy cost, which is 2% of the total. Anyone arguing the hurdle is too high must argue that capex comes in below \$8M per MW-IT sustained, or that accelerators hold value beyond five years; anyone arguing it is too low must argue the reverse. The financing assumptions are the least uncertain input, anchored by KR-WP-2026-10 (Kilambi Research, 2026b).

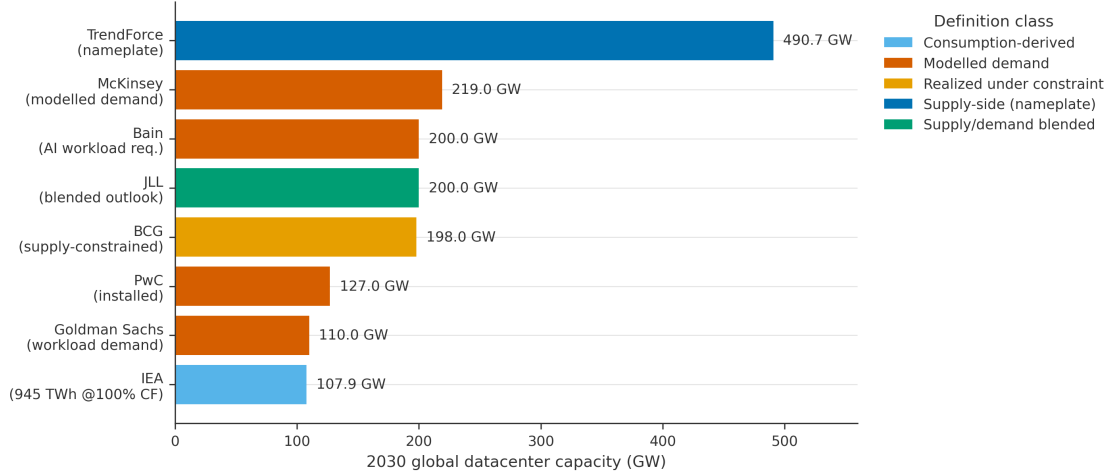


Figure 2. 2030 global datacenter capacity forecasts by definition class. Dispersion is driven by definition: nameplate (TrendForce, 490.7 GW, scope unverified) exceeds modelled demand (McKinsey 219 GW; Bain 200 GW; Goldman Sachs 110 GW), which exceeds installed and supply-constrained figures (PwC 127 GW; BCG 198 GW; JLL 200 GW blended), which exceed consumption-derived figures (IEA 108 GW from 945 TWh at 100% capacity factor). Bars are not comparable across colors.

6.3 The Jevons test

Figure 5 plots the required-capacity response to efficiency gains across the elasticity grid. At the only causal elasticity estimate, $|e| = 0.5$, a 1% efficiency gain reduces required physical capacity, and compounding the token-price efficiency series (161% per year in log terms) predicts required capacity shrinking 55.3% per year (MODELLED, B). The hardware-efficiency series (19% per year) predicts a 9.1% per year shrinkage at the same elasticity. Against this, observed US nominal IT capacity grew 53.1% per year over 2021–2024 (OBSERVED, B). The gap between the predicted -55.3% and the observed $+53.1\%$ must be filled by outward adoption shifts: new workloads, quality expansion, and new users. The required shift is roughly 3.4x per year in multiplicative terms.

The result rejects the price-channel version of the Jevons argument: at inelastic demand, efficiency plus an adoption undershoot is the stranding mechanism, not backfire. If demand were elastic ($|e| > 1$), efficiency would expand required capacity, but the only causal estimate puts demand firmly in the inelastic region, and the EC2 estimate of 1.20 applies to a different market (spot cloud instances, Ben-Yehuda et al. 2013) at a different time. Two caveats apply. First, the elasticity may not be stable: if AI demand becomes more elastic as new use cases arrive, the price channel strengthens. Second, the token-price composite conflates quality change with price change; to the extent that falling token prices reflect better models rather than cheaper compute, the 161% series overstates the effective price decline. Under either caveat, the conclusion is bounded: the price channel as currently measured does not support Jevons backfire, and current capacity growth is an adoption story.

6.4 Scenario revenue requirements

Table 3 and Figure 6 map cumulative 2026–2035 investment into sustained end-market revenue requirements. Each \$1T of cumulative investment requires about \$250B per year of sustained

Table 2. Forecast-audit highlights: the most-quoted figures and what they actually measure. All rows (OBSERVED, B) as records of publication; the underlying forecasts carry the publishers’ own uncertainty, which most do not quantify.

Publisher and figure	What it measures	Definition class
McKinsey, \$6.7T through 2030	Investment <i>including hardware</i> , not capacity	Modelled investment
PwC, \$31.6T through 2050	Investment <i>including repeated hardware refreshes</i>	Modelled investment
JLL, 66 GW under construction	Steel in the ground; 95% precommitted	Under construction
ERCOT, ~410 GW requests	Interconnection requests; 45.4 GW SA-tracked “real”	Announced pipeline
TrendForce, 490.7 GW	Nameplate; scope unverified	Nameplate
IEA, 108 GW (945 TWh at 100% CF)	Consumption-derived capacity equivalent	Consumption-derived
Goldman Sachs, 110 GW	Workload-implied demand	Modelled demand
BCG, 100–130 GWh	Unit error in source; retained flagged	Unclassifiable
Morgan Stanley, 126 GW “annually”	Ambiguous: additions or cumulative	Ambiguous
Dell’Oro, \$3T	Ambiguous: annual or cumulative	Ambiguous

end-market AI revenue in the base case (MODELLED, B). The \$10T base case implies \$2.50T per year supporting 154 GW. The high-demand case, which assumes stronger pricing power, requires less (\$2.18T per year at \$10T). The efficiency/disruption case requires more revenue per GW (because GPU prices fall to \$1.25 per hour) while simultaneously assuming lower utilization and shorter hardware life; required paid utilization reaches 158%, which is arithmetically impossible, so the case is marked not sustainable at every investment level.

Table 3. Cumulative 2026–2035 investment mapped to required sustained end-market AI revenue per year, in \$T. (MODELLED, B) throughout; sustainability means required paid utilization at or below 100%. The efficiency/disruption case (GPU price \$1.25/hour, 45% utilization, four-year life) requires 158% utilization and is unsustainable at every level.

Investment	Base	High demand	Efficiency/disruption
\$5T	\$1.25T	\$1.09T	\$1.42T (not sustainable)
\$10T	\$2.50T	\$2.18T	\$2.83T (not sustainable)
\$15T	\$3.74T	\$3.27T	\$4.25T (not sustainable)

The \$2.50T figure should be read against the size of the AI economy, not against datacenter revenue. End-market AI revenue includes software, services, and productivity value captured by users; Acemoglu (2025) provides the macroeconomic context for judging whether \$2.50T per year is plausible by the early 2030s. The scenario model does not answer that; it prices the claim. Note the linearity: the mapping is roughly proportional because the model holds parameters fixed across investment levels, which understates the likely deterioration of pricing power as capacity grows. The true revenue requirement at \$15T is therefore probably above \$3.74T per year, not equal to it.

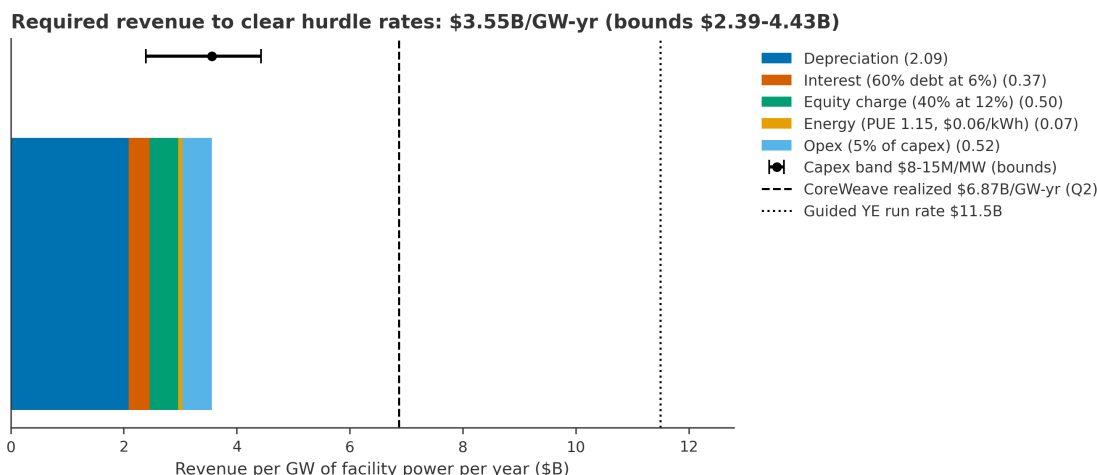


Figure 3. Required revenue per GW of facility power per year. Central estimate \$3.55B (bounds \$2.39B–\$4.43B from capex \$8M–\$15M per MW-IT), decomposed into depreciation (\$2.09B), interest (\$0.37B, 60% debt at 6%), equity charge (\$0.50B, 40% at 12%), energy (\$0.07B, PUE 1.15 at \$0.06/kWh), and opex (\$0.52B, 5% of capex). Dashed and dotted lines show CoreWeave’s Q2 annualized \$6.87B and guided year-end \$11.5B run rates, both reconstructed to per-GW terms.

6.5 Timing mismatch and analogues

Figure 7 shows the two timing calculations. The left panel decomposes 30-year lifetime gross capex per GW: \$180B of accelerator cohorts (six generations at five-year life) against \$20B of facility and power infrastructure. Hardware is 90% of lifetime gross capex but 60% of first cost (MODELLED, B). The right panel shows a single accelerator cohort’s economics: revenue at a 20% annual price decline falls from \$10.65B in year 1 to \$4.37B in year 5, covering only 40% of the fixed \$10.82B annual capital-plus-opex charge by year 5 (MODELLED, B). A 10-year commitment signed at \$2.50 per GPU-hour overpays by 86.8% in present value against a market price falling 15% per year (MODELLED, B).

The mechanism is general: whenever the contracted revenue term exceeds the economic life of the asset generating it, the contract embeds a bet that prices will not fall. In AI compute, prices always fall. The 15% decline rate used for the lease calculation is illustrative; the token-price composite suggests realized declines have been faster, which would make the overpayment larger.

Table 4 reports the three analogues at the investor-return level. In each case the technology succeeded and the original investors lost: Global Crossing’s debt recovered about 5.5 cents on the dollar with equity effectively wiped out; Calpine’s equity fell about 99.4% while senior debt recovered near par; Chinese polysilicon prices fell about 96% and Suntech defaulted. The pattern is consistent with the real-options logic of Dixit and Pindyck (1994): irreversible capacity committed against uncertain demand leaves investors exposed to the full downside of the demand shortfall, and the cost decline reprices the asset base against them.

The analogues discipline the conclusion in both directions. They rule out the claim that overbuilding is harmless because demand always catches up: demand did catch up in all three cases, and investors still lost. They also rule out the claim that overbuilding proves the technology was a mistake: the technology won in all three cases. Applied to AI, the required conclusion is that the technology can succeed while the original investors lose, and the policy-relevant question is who bears the repricing,

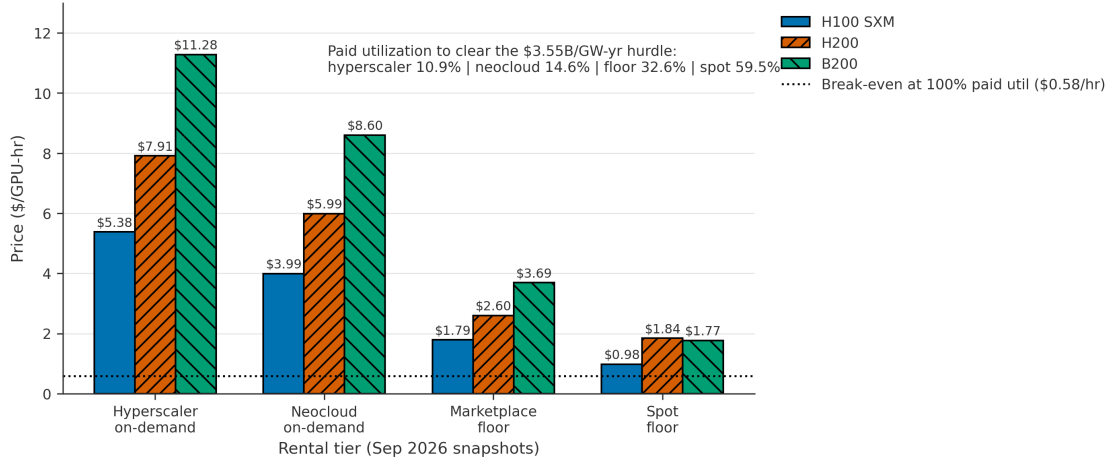


Figure 4. GPU rental tiers (September 2026 snapshots) and the paid utilization required to clear the \$3.55B per GW-year hurdle at each tier’s H100 price: hyperscaler on-demand 10.9%, neocloud on-demand 14.6%, marketplace floor 32.6%, spot floor 59.5%. Dotted line: break-even at 100% paid utilization is \$0.58 per GPU-hour. H200 and B200 tiers shown for context; the hurdle calculation uses H100-class economics.

Table 4. Analogue episodes: technology succeeded, original investors lost. Figures are approximate and drawn from restructuring and market records; see the data appendix for sources. (OBSERVED, B) as historical records.

Episode	Investor-level outcome	Technology outcome
Telecom, late 1990s (Global Crossing)	Debt ~5.5 cents on the dollar; equity effectively wiped out	Fiber demand eventually a
Merchant power, early 2000s (Calpine)	Equity down ~99.4%; senior debt near full recovery	Gas-fired generation remain
Chinese solar, late 2000s (Suntech)	Polysilicon down ~96%; Suntech defaulted	Solar became the cheapest

not whether AI as a technology succeeds.

6.6 Current market signals

Figure 8 codes the 41 market-evidence rows by category and verdict. The pattern is layered, not uniform. Vacancy and preleasing (7 tight) and commitments and orders (7 tight) show a physically tight market for contracted capacity: space is leased before it is built, and equipment orders continue. Interconnection and queues read 1 tight and 3 mixed: the constraint is real but queue reforms and withdrawals muddy the signal. Credit and finance read 3 tight, 4 mixed, and 3 soft: funding is available for contracted projects but spreads and structures are discriminating more than a year ago. Pricing and secondary read 1 tight and 4 soft: marketplace GPU prices and secondary hardware are falling. Cancellations and delays read 2 mixed and 4 soft: uncontracted projects are being deferred or cancelled, including the \$150B-plus of stalled US projects compiled by Finvestly (2026). Utilization and equipment read 1 tight and 1 soft on thin evidence.

The layered verdict is: contracted power, interconnection, land, and preleased capacity remain tight; marketplace GPU prices, secondary hardware, some credit spreads, and uncontracted projects are softening; no verified tenants are walking away from energized capacity. The softness is concentrated

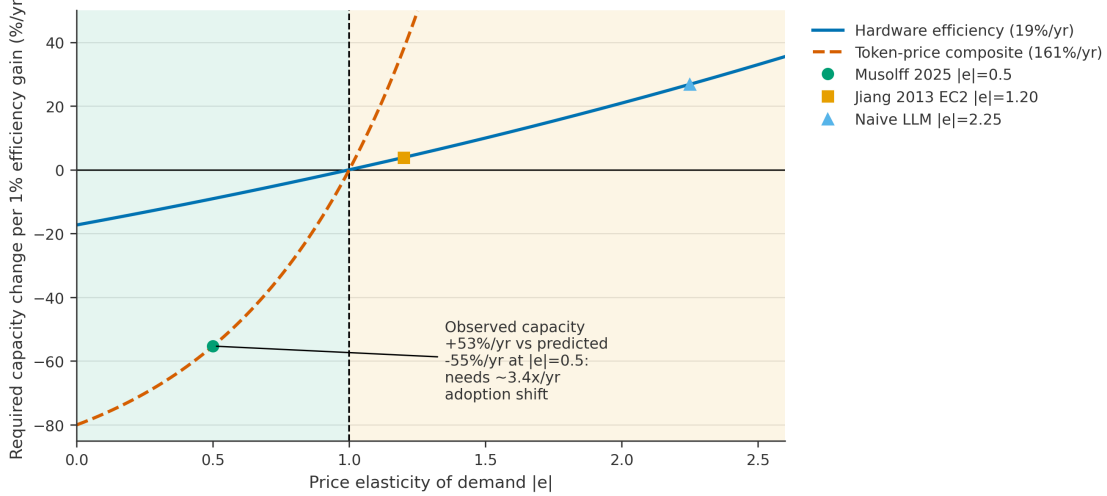


Figure 5. Required capacity change per 1% efficiency gain as a function of the price elasticity of demand, for hardware efficiency (19% per year, solid) and the token-price composite (161% per year in log terms, dashed). Shaded regions: inelastic demand ($|e| < 1$, capacity shrinks) versus elastic demand ($|e| > 1$, Jevons backfire). Markers show literature elasticity points: Musolf 2025 ($|e| = 0.5$), Jiang 2013 on EC2 ($|e| = 1.20$), and a naive LLM-query elasticity ($|e| = 2.25$). At $|e| = 0.5$ the token-price series predicts -55.3% per year against observed $+53.1\%$ per year, requiring roughly 3.4x per year of outward adoption shift.

where it should be if the timing-mismatch mechanism is operating: in the uncontracted, price-exposed layer of the market, not in the contracted base. Direct AI-fleet utilization remains undisclosed by every operator; until it is published, the tightness of the contracted base cannot be distinguished from the tightness of paid demand.

Table 5 organizes the same evidence as an early-warning dashboard: the indicators to watch, the direction that would confirm the stranding mechanism, and the current reading.

6.7 Utilized-compute paths

Figure 9 shows utilized compute under the three demand scenarios on the central capacity anchor, with the shaded band showing the base scenario under the low and high capacity anchors. Utilized compute rises from about 15,500 H100-equivalent EFLOP/s in 2026 to about 48,700 in the 2035 base case and about 74,200 in high adoption; in the disruption case it peaks near 21,000 around 2029 and falls to about 10,600 by 2035, below the 2026 starting level. The capacity-anchor band for the 2035 base case runs from about 29,600 to about 87,000 EFLOP/s: the anchor choice matters, but it does not change the ordering of the scenarios.

Figure 10 translates the same paths into paid utilization against the capital hurdle. The base path clears the 32.6% hurdle (floor price, five-year life) in every year and stays above the 45% line (floor price, three-year life) through 2035, though the margin compresses to about one point by the end of the window. The high-adoption path clears both lines with room to spare. The disruption path falls below the 45% line in 2027 and below the 32.6% line in late 2029; after that the marginal GW does not clear hurdle rates at anything near the floor price. Two qualifications apply. First, the hurdle lines are computed at the September 2026 floor price of \$1.79; in the disruption scenario realized prices fall toward \$0.41 by 2035, which raises the true hurdle, so Figure 10 understates the

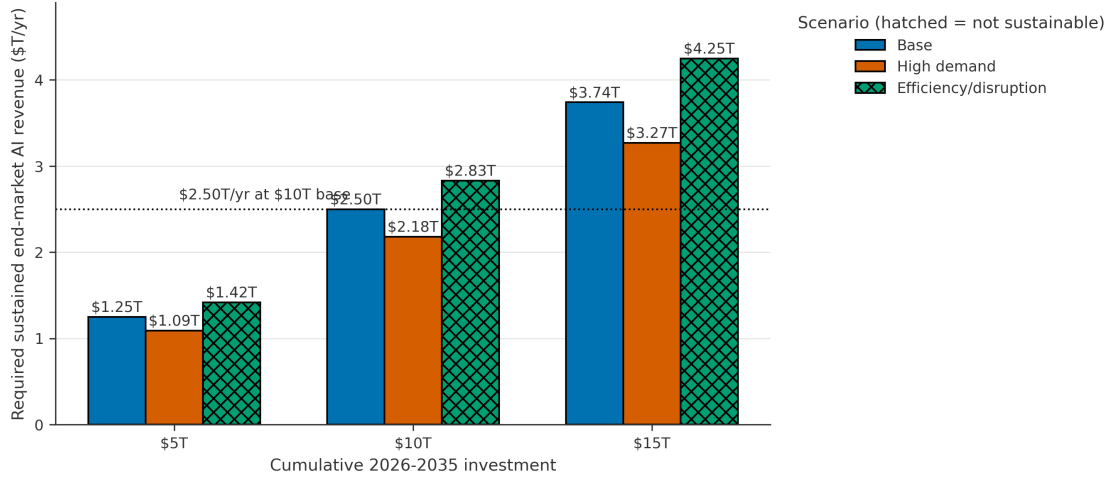


Figure 6. Required sustained end-market AI revenue per year for cumulative 2026–2035 investment of \$5T, \$10T, and \$15T, under base, high-demand, and efficiency/disruption parameter sets. Hatched bars (efficiency/disruption) are not sustainable: required utilization reaches 158%. Dotted line marks the \$2.50T per year required at the \$10T base case.

failure. Second, these are scenario-conditioned paths, not a forecast of what will happen: they show what utilization must look like for each demand regime, given the committed build.

The paths sharpen the paper’s central result rather than changing it. The base case remains absorbable at the floor price, consistent with Section 6.4; the margin of safety is thin and concentrated in the accelerator-life assumption. The disruption case gives a dated form to the stranding mechanism: utilization breaks first, around 2029, and the price collapse follows the utilization break rather than causing it. On the dashboard in Table 5, the indicator to watch is therefore paid utilization itself, if operators ever disclose it: a sustained print below 32.6% at floor pricing while capacity follows the central anchor is the earliest confirmation the disruption path is the one being taken.

6.8 Falsifiers

The paper’s claims are falsifiable. The required-revenue model is falsified if sustained capex below \$8M per MW-IT is documented across multiple AI-ready builds, or if accelerator economic life beyond six years is demonstrated in production fleets. The Jevons result is falsified by a credible causal elasticity estimate above 1.0 for AI compute demand, or by disclosure of AI-fleet utilization showing paid demand absorbing capacity at falling prices without adoption growth. The scenario mapping is falsified if end-market AI revenue reaches \$2.50T per year without the modelled capacity, which would imply the model’s capex-to-revenue linkage is misspecified. The timing-mismatch mechanism is falsified if 10-year fixed-price compute contracts signed in 2024–2025 are still at or above market at expiry, or if compute prices stabilize rather than continuing to fall. The layered market verdict is falsified by verified tenant walkaways from energized capacity (which would show the contracted base is softer than coded) or by a sustained rebound in marketplace pricing (which would show the soft layer was cyclical). The author commits to updating these readings as new data arrive; the dashboard in Table 5 is the instrument for doing so. The utilization paths in Section 6.7 are falsified if disclosed AI-fleet paid utilization diverges persistently from all three paths, for example sustained paid utilization below 30% while capacity follows the central anchor.

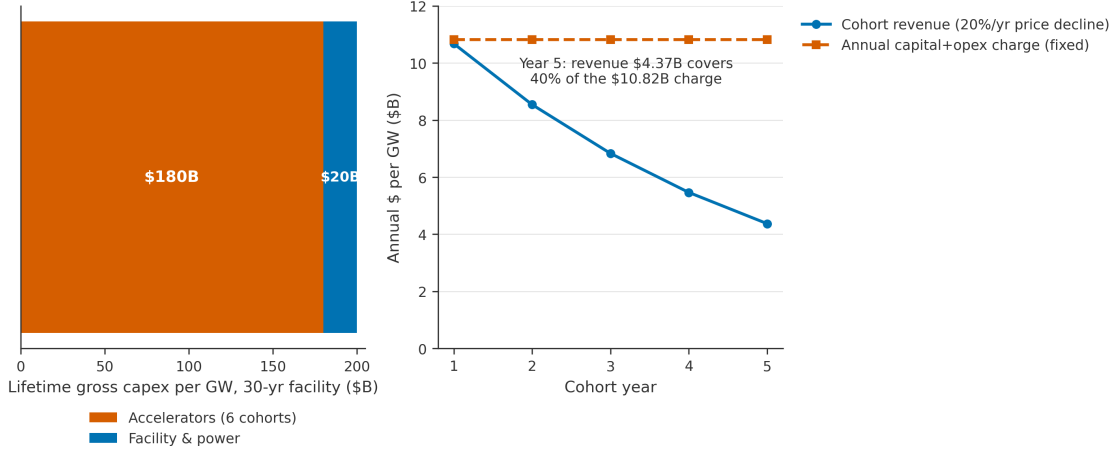


Figure 7. The timing mismatch. Left: 30-year lifetime gross capex per GW is \$200B, of which \$180B (90%) is accelerator cohorts across six five-year generations and \$20B is facility and power; hardware is 60% of first cost. Right: a single cohort’s revenue under 20% annual price declines falls from \$10.65B to \$4.37B over five years against a fixed \$10.82B annual charge, covering 40% of the charge by year 5.

7 Limitations and robustness

Retrospective and non-pre-registered. This paper is retrospective: the datasets were assembled after the buildout was underway, the hypotheses were formed with knowledge of the outcomes, and no analysis plan was pre-registered. The falsifiers in Section 6.8 are stated prospectively to discipline future updates, but they do not cure the retrospective design. Readers should discount the sharpness of the results accordingly.

Forecast audit. The 52-row audit inherits the weaknesses of its sources. Paywalled items (McKinsey, BNEF, Bain, PwC, SemiAnalysis, JPMorgan) are coded from secondary reporting and carry that additional layer of uncertainty. Weak items are flagged rather than dropped, which means the dispersion charts include figures the author would not defend as forecasts. There is no dedicated China forecast workstream; to the extent Chinese capacity follows a different path, the global aggregates are misspecified. The audit counts publications, not independent estimates: several forecasts share data sources and consultants, so 52 rows overstate the effective sample size.

Required-revenue model. The inputs are assumptions, not estimates: capex at \$12M per MW-IT, five-year life, 60/40 financing, PUE 1.15, \$0.06 per kWh, 5% opex. The bounds vary only capex, understating total uncertainty. The model assumes the facility is filled with a single accelerator generation’s economics; in practice, cohorts overlap and newer cohorts have better performance per dollar, which improves the blended economics relative to the single-cohort calculation. Against that, the model assumes no cost overruns and no financing stress, which flatters the economics relative to KR-WP-2026-10’s credit analysis (Kilambi Research, 2026b).

Jevons test. The elasticity grid is illustrative, not estimated: the causal estimate of 0.5 comes from a single study, the EC2 estimate of 1.20 from a different market, and the naive 2.25 is a placeholder. If AI demand is more elastic than 0.5, the price channel is stronger than the central calculation allows. The token-price composite conflates quality and price. The comparison window (2021–2024) is short and covers the post-ChatGPT adoption wave, which biases the observed growth rate upward relative to any steady state.

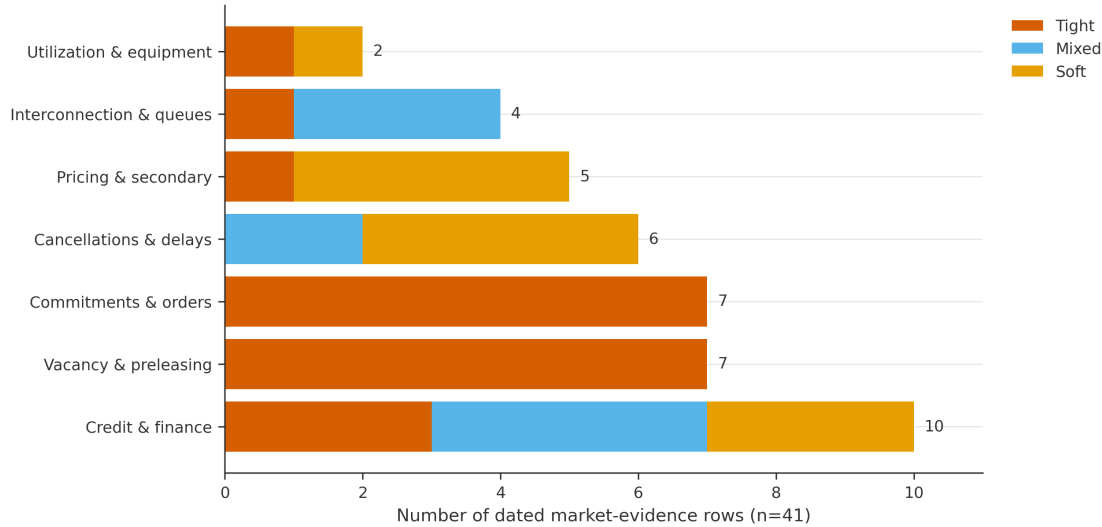


Figure 8. Dated market-evidence rows (n=41) by category and verdict. Contracted capacity (vacancy, commitments) reads tight; marketplace pricing, secondary hardware, and uncontracted projects read soft; credit and interconnection read mixed. Counts are a structured reading of trade press, disclosures, and rating notes, not a statistical sample.

Scenarios. The model is linear in investment, which understates pricing-power deterioration at higher capacity. It ignores financing dynamics, deployment timing, and the distinction between training and inference demand. The efficiency/disruption case's 158% required utilization is an arithmetic result of its inputs, not a prediction; its purpose is to show that the bear case on pricing is incompatible with the bull case on investment, not to forecast utilization.

Utilized-compute forecast. The Section 6.7 paths are constructed, not observed: no public AI-fleet utilization series exists against which to calibrate them, and the 2026 starting stock of about 50 GW is triangulated rather than measured. The capacity path is held scenario-invariant, which understates the true uncertainty if builds are cancelled in the disruption case or accelerated in high adoption; the anchor band is a partial substitute, not a full one. The fixed-silicon EFLOP/s conversion understates effective compute growth from hardware efficiency, which flatters the disruption case's utilized-compute level, while the paid-utilization chart (Figure 10) carries the economic verdict. The paths are scenario conditions, not predictions, and should be read as what each demand regime implies, not as what will happen.

Analogues. Telecom, merchant power, and solar differ from AI datacenters in demand visibility, contract structure, and the pace of cost decline. The analogues discipline the conclusion but do not calibrate it: no quantitative mapping from Global Crossing's 5.5 cents to AI debt recovery is attempted or implied.

Market evidence. The 41 rows are a structured reading, not a sample; the coding reflects author judgment, and the tightness of the contracted base cannot be separated from the tightness of paid demand until utilization is disclosed. The unverified NVIDIA \$99B program, Broadcom \$29B backstop, AMD \$4.1B guarantee, and Goldman 353/381 bp spread figures are excluded and should not be read back into the results.

Table 5. Early-warning dashboard. Each indicator is paired with the reading that would confirm the timing-mismatch stranding mechanism and the current reading from the 41-row dataset. All rows (OBSERVED, B) as records of observation; the “would confirm” column is author judgment.

Indicator	Reading that would confirm stranding	Current reading
Marketplace GPU prices	Sustained fall below \$0.58/hr break-even	Falling; floor \$1.79 (H100)
Secondary accelerator prices	Collapse toward scrap value	Softening
Prelease rates on new builds	Fall below $\sim 80\%$	95% (tight)
Uncontracted project cancellations	Accelerating	Rising (soft)
Tenant defaults on energized capacity	Any verified case	None verified
AI-fleet utilization (if disclosed)	Below 32.6% at floor pricing	Undisclosed
Credit spreads on AI vehicles	Sustained widening	Mixed; discriminating
Interconnection queue withdrawals	Mass withdrawals	Mixed; reforms ongoing

8 Conclusion

The AI buildout faces a demand-absorption question that the supply-side debate has not answered. This paper’s accounting gives the following results. First, published forecasts disagree by roughly a factor of five once definition classes are separated (108 GW consumption-derived to 490.7 GW nameplate, scope unverified), and pipeline figures collapse by 79–89% between interconnection requests and build-ready capacity; no independent source forecasts utilized compute, so Section 6.7 constructs scenario-conditioned paths for it. Second, each GW of facility capacity must generate about \$3.55B per year (bounds \$2.39B to \$4.43B) to clear the capital hurdle, implying 32.6% paid utilization at the marketplace floor price of \$1.79 per GPU-hour, rising to 45% at a three-year hardware life; all figures (MODELLED, B) from stated inputs. Third, the Jevons argument fails the price-channel test: at the only causal elasticity estimate, efficiency predicts required capacity shrinking 55% per year against observed growth of 53% per year, so current growth is an adoption story. Fourth, each \$1T of cumulative 2026–2035 investment requires about \$250B per year of sustained end-market AI revenue in the base case (\$2.50T per year at the \$10T base case, supporting 154 GW); the efficiency/disruption case is arithmetically unsustainable at 158% required utilization. Fifth, the central stranding mechanism is the timing mismatch: hardware is 90% of 30-year lifetime capex but 60% of first cost, and long-dated fixed-price commitments overpay by 87% in present value if prices fall 15% per year. Sixth, the analogues show technology can succeed while original investors lose. Seventh, current market signals are layered: contracted capacity tight, price-exposed layers softening, no verified walkaways from energized capacity, and AI-fleet utilization undisclosed.

The paper issues no binary bubble verdict, because the evidence does not support one. What it supports is a narrower claim: the investment case depends on adoption growth continuing at roughly its 2021–2024 pace, on pricing power holding at tiers well above spot, and on contract structures that do not yet price the decline curve of the assets they are written against. Each of those is observable. Table 5 is the instrument for watching them.

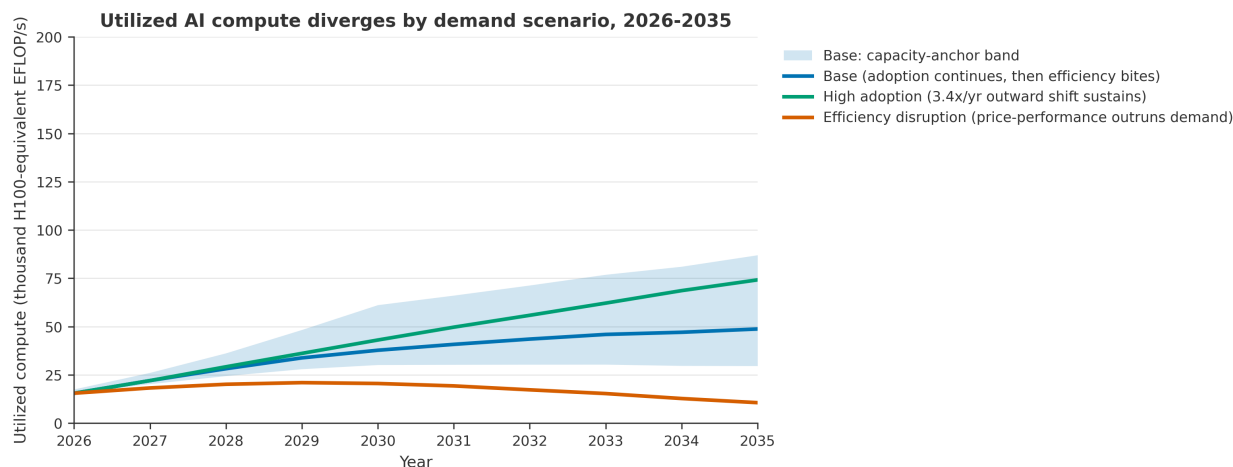


Figure 9. Utilized AI compute, 2026–2035, under three demand scenarios on the central capacity anchor (50 GW in 2026 rising linearly to 154 GW in 2035). Shaded band: the base scenario under the low and high capacity anchors. Units are H100-equivalent EFLOP/s at 2026 silicon (688 EFLOP/s per GW at full utilization). (MODELLED, C) throughout: these are scenario-conditioned paths, not a forecast of what will happen.

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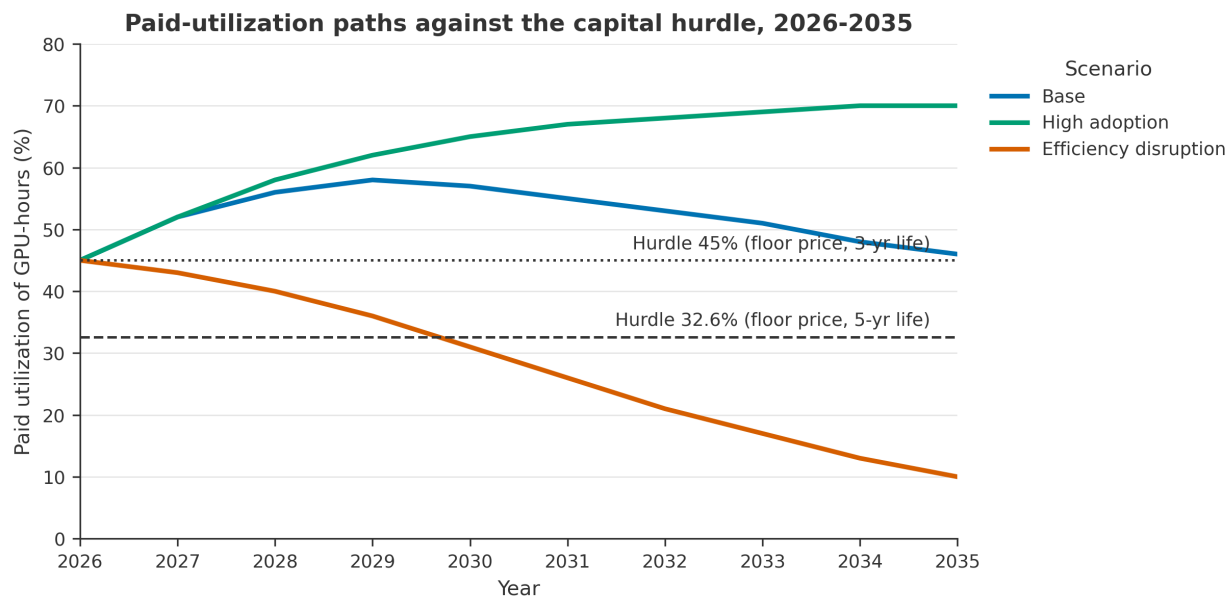


Figure 10. Paid-utilization paths, 2026–2035, against the capital hurdle. Dashed and dotted lines: the required-revenue model’s break-evens at the September 2026 marketplace floor price of \$1.79 per GPU-hour, 32.6% at a five-year accelerator life and 45% at a three-year life (MODELLED, B). Utilization paths (MODELLED, C). In the disruption scenario the true hurdle rises as realized prices fall, so the chart understates the failure.

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A Data appendix

Files. The analysis uses five CSV datasets stored in the project `data/` directory: `forecast_audit.csv` (52 rows), `demand_chain.csv`, `market_evidence.csv` (41 rows), `scenario_outputs.csv`, and `utilization_forecast.csv` (10 rows: capacity anchors, utilization and price paths, utilized EFLOP/s, hurdle lines). Figures 1–8 are generated by `scripts/make_figs.py` and Figures 9–10 by `scripts/utilization_forecast.py` from these files; no figure contains hand-entered data.

Forecast audit variables. `publisher`, `pub_date`, `target_year`, `headline_figure`, `units`, `definition_class` (one of modelled demand, announced pipeline, under construction, contracted, energized, nameplate, utilized), `scope_note` (e.g., “including hardware,” “including refreshes”), `weakness_flag` (unit problem, ambiguity, paywalled/secondary-only, scope unverified), `evidence_grade`.

Demand-chain variables. Each row is one link: `link_name`, `value`, `units`, `year`, `provenance`, `evidence_grade`. Links include the funnel ratios, the required-revenue components and inputs, the GPU tier price snapshots (September 2026), the Jevons elasticity grid and efficiency series, the scenario-model parameters, and the timing-mismatch parameters.

Market-evidence variables. `date`, `observation`, `category` (one of the seven listed in Section 5.8), `verdict` (tight, mixed, soft), `source`, `evidence_grade`.

Scenario-model variables. `investment_level` (\$5T/\$10T/\$15T), `parameter_set` (base, high demand, efficiency/disruption), `required_revenue_per_yr`, `implied_gw`, `required_utilization`, `sustainable` (yes/no).

Utilization-forecast variables. `year` (2026–2035), `capacity_central_GW`, `capacity_low_GW`, `capacity_high_GW` (the three capacity anchors), `util_base_pct`, `util_high_pct`, `util_disruption_pct` (paid-utilization paths), `price_base_usd`, `price_high_usd`, `price_disruption_usd` (realized \$/GPU-hour by scenario), `eflops_base`, `eflops_high`, `eflops_disruption` (utilized H100-equivalent EFLOP/s on the central anchor), `eflops_base_low_anchor`, `eflops_base_high_anchor` (base path on the low and high anchors), `hurdle_5yr_pct` (32.6), `hurdle_3yr_pct` (45.0), `grade`.

Analogue sources. Telecom: Global Crossing bankruptcy and recovery records; merchant power: Calpine restructuring records; solar: polysilicon price series and Suntech default records. Figures are approximate; the paper uses them as order-of-magnitude discipline, not calibration.

Production note. This paper was produced on 2026-09-29 using subagents and AI research assistance. Quantitative analysis was checked against the source datasets; every number in the text traces to a dataset row or a stated model input. The author reviewed the manuscript for consistency with the evidence, including findings adverse to the investment case. No part of the analysis was pre-registered. Section 6.7 and Figures 9–10 (the utilized-compute forecast) were added after the paper’s initial release on 2026-09-29, at the author’s direction; the paper number and title are unchanged.